Hans Martinez

Ph.D. Candidate in Economics

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I am an applied microeconomist specializing in industrial organization. A distinctive feature of my recent research is that it uses economic theory to model the role of unobservables in the data-generating process. Two threads play a prominent role in my research: strategic misconduct and an interest in how market structure and its institutional environments shape agents' incentives.

RESEARCH INTERESTS

Fields: Industrial Organization, Applied Microeconomics
Subfields: Empirical Industrial Organization, Productivity and Production-Function Estimation, Market Power and Markups, Tax Evasion and Compliance

EDUCATION

Ph.D. in Economics, Candidate, Western University *2019 – Present*
- Job Market Paper: *Tax Evasion and Productivity*
- Supervisor: Salvador Navarro
- Committee Members: David Rivers, Timothy Conley

M.S. in Economics and Industrial Organization, Faculty of Economics, UANL *2014 – 2016*
- Thesis: *e-Rational Voters: Measuring the Effects of Internet and Media Freedom on the Political Budget Cycle in a Weak Democracy*

WORKING PAPERS

Tax Evasion and Productivity. (Job Market Paper)

Cournot Profitability and Measurement Error. (Summer Paper)

WORK IN PROGRESS

The Economic Impact of Community Resource Management Areas in Ghana. (with Timothy Conley, Isaac Luginaah, and Cornelius Pienaah).

Generative AI and Worker Productivity: A Field Experiment. (with Martin Petrin).

GenAI and Productivity in Canada: A Production Function Approach.

Inference for Gross-Output Production Functions under Cross-Firm Dependence.

Identification of Latent Cost Functions in Homogeneous Goods Markets.

PUBLICATIONS

Is Internet Access Bad News for Media-Capturing Incumbents? (with C. Gámez), *Journal of Applied Economics*, 2019. Vol. 22, No. 1, pp. 527-553.

REPORTS

Analysis of Risks and Opportunities of Rare Earth Elements for Financial Institutions in Canada. (with Javier Martinez Hernandez), *Global Risk Institute*, 2026.

TEACHING EXPERIENCE

Western University — London, ON

Financial Economics Refresher, *Master in Financial Economics*	*Summer 2025 and 2026*
Intermediate Microeconomic Theory I: ECON2150	*Fall 2025*
Intermediate Microeconomic Theory II: ECON2151	*Winter 2024*
Public Finance - Taxation: ECON3329	*Fall 2024*
Econometrics I: ECON2122	*Summer 2022 and Fall 2024*

RESEARCH EXPERIENCE

Western University — London, ON

Graduate Research Assistant — *2020 – 2026*

- Research assistant for multiple professors besides my supervisors over the course of the Ph.D., including Juan Carlos Hatchondo, Martin Petrin, Christian Dippel, and Daniel Chaves.

Brock University — St. Catharines, ON

Research Assistant to Andrew Dickens — *2020 – 2021*

Economics Research Centre, Faculty of Economics, UANL — Monterrey, MX

Economics Journal Coordinator — *2016 – 2023*

- *Ensayos, Revista de Economía*, Faculty of Economics at UANL

SKILLS

Software: R, Julia, Stata, Fortran, Matlab
Languages: English (Fluent), Spanish (Native), French (Intermediate)

REFERENCES

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ABSTRACTS

Tax Evasion and Productivity (Job Market Paper)

Value-added taxes (VAT) allow firms to deduct the taxes paid on purchases from the taxes collected on sales, creating an incentive for firms to overreport the purchases of inputs to reduce their tax bill. The fundamental issue in detecting input overreporting stems from two unobservables, true inputs and productivity. Because firms differ in productivity, a firm reporting high input use, for a given level of output, may be either overreporting or simply less productive. To distinguish between overreporting and productivity, I use a structural production-function approach and a benchmark group, a subset of firms that report inputs correctly. Because the benchmark firms are assumed to share the same technology as the other firms, I estimate the technology's parameters from them. Using these estimates, I compare what potentially tax-evading firms report with predictions of the true inputs they would use. I apply the method to a Colombian manufacturing survey from 1981 to 1991. During this period, the Colombian sales tax worked as a VAT and the data include a subset of firms that, due to government and market scrutiny, reported inputs correctly.

My approach yields a simple and robust statistical test to detect input overreporting. Under the null hypothesis of correct reporting, the test is agnostic about the specification of both the probability of detection and the cost of overreporting. Among the 5 largest industries, the test rejects the null hypothesis in all 4 industries subject to the sales tax, and, as expected, it fails to reject in the industry exempt from the sales tax. I recover the distribution of overreporting by deconvolution: in the five largest of the industries where the test rejects at the 1 percent level and the deconvolution applies, firms overreport on average between 4.5 and 23.1 percent of their true inputs, depending on the industry. Exploiting a 1983 fiscal reform in Colombia that raised the sales tax rate, I then document that firms overreport more after the increase. The response implies a midpoint elasticity of overreporting with respect to the tax rate of 5.9.

The magnitude of detected overreporting matters for methods of estimating production functions that exploit a flexible input to identify productivity (Levinsohn and Petrin 2003; Doraszelski and Jaumandreu 2013; Ackerberg et al. 2015; Gandhi et al. 2020). These methods typically assume that materials are flexible, and their demand or first-order conditions reveal productivity. However, firms overreport materials. I show that failing to correct for overreporting yields larger output elasticities of materials, and more dispersed and less persistent productivity distributions. Using the production-function parameters, I estimate a structural model of input overreporting. With the model, I study how net sales-tax revenue changes as claimed deductions, true and fictitious, respond to the sales-tax rate on purchases. At the current rate, a 1 percent increase in the rate lowers net sales-tax revenue by about 1.7 percent of gross sales-tax revenue. Undetected overreporting costs the government 4.7 percent of the gross sales-tax revenue from potentially tax-evading firms.

Cournot Profitability and Measurement Error (Summer Paper)

This paper tests whether firms compete by choosing quantities (Cournot competition) when quantities are measured with error. I extend Aguiar and Kashaev (2020), who introduced measurement error into the Revealed Preference test (Afriat 1967), to the deterministic *Revealed Profitability* (RP) test of the Cournot model (Carvajal et al. 2013). The RP analysis derives a set of inequalities from the convexity of the firm's profit-maximization problem. The convexity stems from the demand function and the firm's cost function, which are both unknown, but the test does not require any parametric assumptions on either. Under the null hypothesis, firms compete in quantities, but observed quantities are mismeasured. To address the measurement error, the test needs only a

centering condition, which fixes the location of the true quantities. If the test fails to reject the null hypothesis, it can then be inverted to answer counterfactual questions, such as how much firms would adjust their output in response to a change in price. I apply the method to the international crude oil market, where measurement error might arise from meter and gauge calibration, human recording errors, and the conversion of raw volumes to standard barrels. As the centering condition, I assume measured quantities are correct on average over each half-year. Unlike the deterministic test, the test allowing for measurement error no longer rejects the null hypothesis of Cournot competition at the 5 percent significance level. In this application, failing to account for measurement error thus leads to an over-rejection of the Cournot model.